

FAKE NEWS DETECTION USING MACHINE LEARNING

Cover Page

Project Title: Fake News Detection Using Machine Learning

Submitted By: MASEERA KOWSAR M

Internship: Artificial Intelligence

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1. Introduction

Fake news refers to false or misleading information presented as genuine news. Due to the rapid growth of social media and online platforms, fake news spreads quickly and influences public opinion. Detecting fake news manually is difficult because of the large volume of information available online.

Machine Learning and Natural Language Processing (NLP) can help automatically identify fake news articles by analyzing their text content. This project aims to develop a machine learning model that can classify news articles as fake or real.

2. Problem Statement

The spread of fake news has become a major issue worldwide. It can mislead people, create confusion, and negatively impact society. Therefore, an automated system is needed to detect fake news efficiently and accurately.

3. Objectives

- To collect and analyze news datasets.
- To preprocess textual data.
- To convert text into machine-readable format.
- To train a machine learning model.

- To classify news articles as fake or real.
 - To evaluate model performance.
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4. Dataset Description

Dataset Name:

Fake and Real News Dataset

Files Used:

- Fake.csv
- True.csv

Dataset Features:

- Title
- Text
- Subject
- Date

The dataset contains both fake and real news articles which are used to train and test the machine learning model.

5. Tools and Technologies Used

Programming Language:

- Python

Libraries:

- Pandas
- NumPy
- Scikit-learn
- NLTK
- Matplotlib
- Seaborn

- WordCloud

Development Environment:

- Visual Studio Code
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6. Methodology

Step 1: Data Collection

The dataset was downloaded and loaded into Python.

Step 2: Data Cleaning

Missing values and duplicate records were removed.

Step 3: Text Preprocessing

Text data was cleaned and prepared for machine learning.

Step 4: Feature Extraction

TF-IDF Vectorization was used to convert text into numerical features.

Step 5: Model Training

A machine learning classifier was trained using the processed data.

Step 6: Model Evaluation

The model performance was evaluated using accuracy metrics.

7. Exploratory Data Analysis

Various visualizations were generated to understand the dataset.

Examples:

- Fake vs Real News Distribution
- Word Cloud of Fake News
- Word Cloud of Real News
- Most Frequent Words

(Insert screenshots here)

8. Results

The machine learning model successfully classified news articles into fake and real categories.

Model Used:

Passive Aggressive Classifier

Accuracy:

(Write your actual accuracy value here)

The model achieved high accuracy and demonstrated effective fake news classification.

9. Conclusion

This project demonstrates how Machine Learning and NLP techniques can be used to detect fake news automatically. The developed model can help reduce misinformation and improve the reliability of online information sources.

10. Future Scope

- Use Deep Learning models for better accuracy.
 - Develop a web application for real-time fake news detection.
 - Integrate social media monitoring features.
 - Expand the dataset for improved performance.
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11. References

- Kaggle Fake and Real News Dataset
- Python Documentation
- Scikit-learn Documentation
- NLTK Documentation